

AI LEGAL™

AI Mock Courtroom Simulator System

Virtual Court Proceedings • Multi-Jurisdiction Simulation • Opposing Counsel Objections • Bench Analytics

Document Area	Details
Product	AI LEGAL™ AI Mock Courtroom
Primary Workspace	MockCourtroomWorkspace.jsx
Core Scope	Realistic oral-argument and courtroom simulation
Key Capabilities	Judicial personas, opposing counsel, objections, timed hearings, transcript and performance analytics

🏛️ AI LEGAL — AI Mock Courtroom Simulator System

Virtual Court Proceedings, Multi-Jurisdiction Simulation, Opposing Counsel Objections & Bench Analytics

1. Executive Summary & Core Objective

Arguing before a judicial bench requires quick thinking, deep familiarity with the evidentiary record, adherence to court decorum, and the ability to counter unexpected legal objections from opposing counsel. Traditionally, advocates had no safe environment to practice oral arguments before stepping into a High Court or Supreme Court hearing.

The \\AI LEGAL AI Mock Courtroom\\ (/dashboard/tools/mock-courtroom\\ / \\MockCourtroomWorkspace.jsx) is a realistic court simulation environment powered by generative legal intelligence. It reproduces the pressure and dynamics of a live trial.

Primary Capabilities:

- \\Dual-Jurisdiction Support:\\ Dedicated simulation modes for \\India\\ (District Courts, High Courts, Supreme Court of India, NCLT/NCDRC) and \\Nepal\\ (सुप्रीम कोर्ट, उच्च न्यायालय, जिल्ला न्यायालय, न्यायाधीश, न्यायाधीश्वर).
- \\Multi-Character Trial Ecosystem:\\ Simulates a Presiding Judge, an aggressive Opposing Senior Counsel, a Key Witness, and a Court Clerk.
- \\Dynamic Judicial Temperaments:\\ Choose between \\Standard Bench\\, \\Moderate Bench\\, or \\Strict / Oppositional Bench\\ to test readiness against tough questioning.
- \\6 Realistic Trial Phases:\\ Practice \\Opening Statements\\, \\Evidence Presentation\\, \\Witness Examination\\, \\Cross-Examination\\, \\Final Arguments\\, or run a \\Complete Trial\\.
- \\Interactive Voice & Speech Simulation:\\ Hands-free voice argumentation using Web Speech API (Speech-to-Text) and natural Text-to-Speech (TTS) judicial audio feedback.
- \\Comprehensive Post-Trial Analytics:\\ Generates a judicial scorecard assessing argument strength, procedural compliance, evidentiary soundness, and citation accuracy.

2. Technical Architecture & Simulation Engine

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Start[Advocate selects Case or enters Custom Brief] --> Setup[Jurisdiction, Court, Bench & Style Setup]

Setup --> Launch[Initialize Courtroom Session]

Launch --> Characters[Multi-Character Simulation Engine]

subgraph "Simulated Courtroom Characters"

Clerk[Court Clerk: Announces matter & enforces decorum]

Judge[Presiding Judge: Queries law, asks clarifications, rules on objections]

Opponent[Opposing Counsel: Interposes timely legal objections]

Witness[Witness in Box: Testifies under examination]

end

Advocate[Advocate User: Voice Input / Text Arguments] --> Characters

Characters --> Gemini[Gemini Pro Legal Reasoning Service]

Gemini --> AudioEngine[Web Speech STT / TTS Audio Engine]

AudioEngine --> Advocate

SessionEnd[Conclude Arguments / Adjournment] --> Evaluation[AI Judicial Evaluation Engine]

Evaluation --> Report[Performance Scorecard & Bench Analytics]

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Key Source Files:

- \\Frontend Workspace:\\ \\d:\\AI
Legal\\AILEgalApp-backend-webpage-\\AI-LegalAppWebapp\\src\\pages\\MockCourtroomWorkspace.jsx\\

- \\Generative AI Service:\\ \\d:\\AI
Legal\\AILEgalApp-backend-webpage-\\AI-LegalAppWebapp\\src\\services\\geminiService.js\\

- \\Case Roster Connector:\\ \\d:\\AI
Legal\\AILEgalApp-backend-webpage-\\AI-LegalAppWebapp\\src\\services\\apiService.js\\

3. Jurisdictions, Court Hierarchies & Bench Difficulties

3.1 Court Selections by Legal Jurisdiction

The simulator automatically detects the advocate's legal jurisdiction and tailors procedural rules accordingly:

Indian Legal System:

1. \\District & Sessions Court:\\ Focus on evidentiary proof, framing of charges, examination under Section 313 CrPC / BNSS, and civil trial procedure (CPC).
2. \\High Court of Judicature:\\ Focus on writ jurisdiction (Articles 226/227), second appeals, revision petitions, and constitutional questions.
3. \\Supreme Court of India:\\ Special Leave Petitions (SLPs under Article 136), appellate scrutiny, substantial questions of law, and binding constitutional bench ratios.
4. \\Commercial Appellate Tribunals (NCLT / NCDRC / NCLAT):\\ Insolvency & Bankruptcy Code (IBC 2016), commercial dispute speed trials, and consumer protection acts.

Nepalese Legal System (नेपाली न्यायिक प्रणाली):

1. \\District Court (जिल्ला न्यायालय):\\ First instance civil and criminal cases under the Muluki Ain / Muluki Criminal & Civil Codes 2074.
2. \\High Court Patan / Biratnagar (उच्च न्यायालय):\\ First appeals, writ petitions, and supervisory review.
3. \\Supreme Court of Nepal (सुप्रीम कोर्ट):\\ Constitutional bench hearings, extraordinary jurisdiction, and precedent formulation.
4. \\Debt Recovery Tribunal / Special Court (वित्त प्रतिकार न्यायालय):\\ Banking recovery and anti-corruption trials under the Special Court Act 2059.

3.2 Bench Temperament & Difficulty Levels

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// MockCourtroomWorkspace.jsx (Lines 28-32)

const DIFFICULTIES = [

{ id: 'Standard', label: 'Standard Bench', desc: 'Normal judicial questioning and procedural compliance' },
 { id: 'Moderate', label: 'Moderate Bench', desc: 'Probing statutory inquiries and evidence verification' },
 { id: 'Strict', label: 'Strict / Oppositional Bench', desc: 'Aggressive questioning, frequent opponent objections & strict ratios' }

];

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- \\Standard Bench:\\ Courteous judicial bench. Gives the advocate ample time to state the facts before asking questions on limitation and maintainability.

- \\Moderate Bench:\\ Active judicial inquiry. Interrupts oral submissions when citations are incomplete or when assertions stray from the pleadings.

- \\Strict / Oppositional Bench:\\ Challenging simulation. The judge frequently presses for immediate statutory authority, demands direct answers, and entertains sharp objections from opposing counsel.

4. The 6 Trial Styles & Practical Applications

Advocates can choose to practice a specific segment of trial advocacy:

\\ Trial Style | Focus Area | Educational & Practical Value |

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\\ \\Complete Trial\\ | Full trial simulation from clerk announcement to judgment reserved. | Comprehensive preparation for final hearings and trials. |

\\ \\Opening Statement\\ | Setting out facts, timeline of injury/breach, and establishing legal theory of the case. | Helps junior advocates master concise, persuasive introductory submissions. |

\\ \\Evidence Presentation\\ | Proving admissibility of documents, electronic records certificates (Section 65B IEA / Section 63 BSA). | Overcoming objections regarding relevance, hearsay, and primary vs secondary evidence. |

\\ \\Witness Examination\\ | Examination-in-Chief of prosecution or plaintiff witnesses. | Formulating non-leading questions that allow the witness to present testimony naturally. |

\\ \\Cross-Examination\\ | Confronting hostile or opposing witnesses with prior contradictions, FIR statements, and email trails. | Mastering leading questions, testing witness credibility, and exposing discrepancies. |

\\ \\Final Arguments\\ | Summing up the evidence, establishing legal liability, addressing precedent citations, and praying for specific relief. | Perfecting appellate submissions and high-court oral advocacy. |

5. Multi-Character AI Simulation in Action

During a live simulation, four distinct courtroom personas interact dynamically:

Character Roles:

1. \\The Court Clerk (\\currentSpeaker: 'clerk'):\\

- \\\"Item Number 14 on the cause list: Commercial Suit 102/2026. M/s Skylark Logistics versus State of Delhi. Counsel for the Petitioner, you may proceed.\"\\

2. \\The Presiding Judge (\\currentSpeaker: 'judge'):\\

- Intervenes to query maintainability, verify statutory thresholds, and rule on objections.
- \\\"Counsel, before you delve into the merits, please satisfy this bench on how your writ petition is maintainable when an efficacious alternate remedy exists under Section 11 of the Arbitration Act?\"\\

3. \\The Opposing Senior Counsel (\\currentSpeaker: 'opponent'):\\

- Interposes procedural and substantive objections in real time:
- \\\"Objection, My Lord! The learned counsel is referring to an uncertified photocopy of an email trail. This violates the mandatory requirements of electronic evidence certification!\"\\

4. \\The Witness (\\currentSpeaker: 'witness'):\\

- Answers direct questions and responds defensively or nervously during cross-examination.

6. Hands-Free Voice Arguments (Web Speech STT & TTS)

The workspace provides full interactive voice capabilities:

- \\Speech-to-Text (STT):\\ Advocates can speak naturally into their microphone. The platform transcribes oral arguments into text in real time.

- \\Text-to-Speech (TTS):\\ The simulated Judge and Opposing Counsel speak their replies using distinct audio voices, allowing the advocate to practice eyes-up oral advocacy without constantly staring at a keyboard.

- \\Mute / Pause Controls:\\ Advocates can pause the proceedings (\\isHearingPaused) to look up an authority or mute TTS audio during quiet study sessions.

7. Post-Trial Judicial Evaluation Scorecard

At the conclusion of the session, the platform runs an in-depth evaluation across five core trial advocacy metrics:

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pie title Post-Trial Judicial Evaluation Scorecard (100 Point Scale)

"Argument Strength & Legal Theory" : 25

"Procedural & Evidentiary Compliance" : 25

"Reaction to Opposing Objections" : 20

"Citation Accuracy & Precedent Handling" : 15

"Court Decorum & Articulation" : 15

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Evaluation Criteria:

1. \\Argument Strength (25 Pts):\\ How effectively the advocate linked statutory provisions to the undisputed facts of the case.

2. \\Procedural & Evidentiary Soundness (25 Pts):\\ Proper handling of exhibit references, burden of proof (Sections 101–103), and electronic record certificates.

3. \\Handling Objections (20 Pts):\\ How quickly and persuasively the advocate countered objections from the opposing counsel.

4. \\Citation Accuracy (15 Pts):\\ Precision in citing Supreme Court and High Court authorities without misrepresenting ratios.

5. \\Decorum & Articulation (15 Pts):\\ Adherence to courtroom formality, respectful address ("My Lord", "Learned Senior Counsel"), and clarity of expression.

The advocate receives a comprehensive PDF report with inline judicial feedback and specific suggestions for improvement before their actual court appearance.